

```
In [1]: import sqlite3
from datetime import datetime

# Connect to SQLite database (creates it if it doesn't exist)
conn = sqlite3.connect('sales_data.db')
cursor = conn.cursor()

# Create the sales table
cursor.execute('''
CREATE TABLE IF NOT EXISTS sales (
    product_id INTEGER,
    product_name TEXT,
    order_date TEXT,
    qty INTEGER,
    price REAL,
    sales REAL
)
''')

# Sample data (10 rows)
sample_data = [
    (1, 'Widget A', '2025-04-21', 3, 19.99, 59.97),
    (2, 'Widget B', '2025-04-22', 1, 29.99, 29.99),
    (3, 'Gadget C', '2025-04-23', 2, 14.99, 29.98),
    (4, 'Gadget D', '2025-04-24', 5, 9.99, 49.95),
    (5, 'Tool E', '2025-04-25', 4, 24.99, 99.96),
    (6, 'Tool F', '2025-04-26', 2, 49.99, 99.98),
    (7, 'Device G', '2025-04-27', 1, 199.99, 199.99),
    (8, 'Device H', '2025-04-28', 3, 89.99, 269.97),
    (9, 'Widget I', '2025-04-29', 6, 12.99, 77.94),
    (10, 'Gadget J', '2025-04-30', 7, 8.99, 62.93),
]

# Insert sample data
cursor.executemany('''
INSERT INTO sales (product_id, product_name, order_date, qty, price, sales)
VALUES (?, ?, ?, ?, ?, ?)
''', sample_data)

# Commit and close
conn.commit()
conn.close()

print("Database 'sales_data.db' with 'sales' table created successfully.")

Database 'sales_data.db' with 'sales' table created successfully.
```

```
In [7]: !pip install ipython-sql

Requirement already satisfied: ipython-sql in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (0.5.0)
Requirement already satisfied: prettytable in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from ipython-sql) (3.16.0)
Requirement already satisfied: ipython in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from ipython-sql) (9.2.0)
Requirement already satisfied: sqlalchemy>=2.0 in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from ipython-sql) (2.0.40)
Requirement already satisfied: sqlparse in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from ipython-sql) (0.5.3)
Requirement already satisfied: six in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from ipython-sql) (1.17.0)
Requirement already satisfied: greenlet<=1 in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from sqlalchemy>=2.0->ipython-sql) (3.2.1)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from sqlalchemy>=2.0->ipython-sql) (4.13.0)
Requirement already satisfied: colorama in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from ipython->ipython-sql) (0.4.6)
Requirement already satisfied: decorator in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from ipython->ipython-sql) (5.2.1)
Requirement already satisfied: ipython-pygments-lexers in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from ipython->ipython-sql) (1.1.1)
Requirement already satisfied: pygments>=2.4.0 in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from ipython->ipython-sql) (0.19.2)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from ipython->ipython-sql) (3.0.51)
Requirement already satisfied: jedi>=0.16 in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from ipython->ipython-sql) (0.19.1)
Requirement already satisfied: stack_data in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from ipython->ipython-sql) (0.6.3)
Requirement already satisfied: traitlets>=6.13.0 in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from ipython->ipython-sql) (5.14.3)
Requirement already satisfied: wcwidth in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from prompt_toolkit<3.1.0,>=3.0.41->ipython->ipython-sql) (0.2.13)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from jedi>=0.16->ipython->ipython-sql) (0.8.4)
Requirement already satisfied: executing>=1.2.0 in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from stack_data->ipython->ipython-sql) (2.2.0)
Requirement already satisfied: asttokens>=1.0.0 in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from stack_data->ipython->ipython-sql) (3.0.0)
Requirement already satisfied: pure-eval in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from stack_data->ipython->ipython-sql) (0.2.3)
```

```
In [8]: %load_ext sql
%sql sqlite:///sales_data.db

The sql extension is already loaded. To reload it, use:
%reload_ext sql

In [19]: !pip install prettytable==3.7.0

Collecting prettytable==3.7.0
  Downloading prettytable==3.7.0-py3-none-any.whl.metadata (26 kB)
Requirement already satisfied: colorama in c:\users\sowriatwork\appdata\local\programs\python\python313\lib\site-packages (from prettytable==3.7.0) (0.2.13)
  Downloading prettytable==3.7.0-py3-none-any.whl (27 kB)
Installing collected packages: prettytable
  Attempting uninstall: prettytable
    Found existing installation: prettytable 3.16.0
    Uninstalling prettytable==3.16.0:
      Successfully uninstalled prettytable==3.16.0
  Successfully installed prettytable==3.7.0
```

```
In [20]: %load_ext sql
%sql sqlite:///sales_data.db

The sql extension is already loaded. To reload it, use:
%reload_ext sql

In [21]: %reload_ext sql

In [25]: import sqlite3
import pandas as pd

conn = sqlite3.connect('sales_data.db')
df = pd.read_sql_query('SELECT * FROM sales', conn)
df.head()
```

```
Out[25]:
```

	product_id	product_name	order_date	qty	price	sales
0	1	Widget A	2025-04-21	3	19.99	59.97
1	2	Widget B	2025-04-22	1	29.99	29.99
2	3	Gadget C	2025-04-23	2	14.99	29.98
3	4	Gadget D	2025-04-24	5	9.99	49.95
4	5	Tool E	2025-04-25	4	24.99	99.96

```
In [27]: df
```

```
Out[27]:
```

	product_id	product_name	order_date	qty	price	sales
0	1	Widget A	2025-04-21	3	19.99	59.97
1	2	Widget B	2025-04-22	1	29.99	29.99
2	3	Gadget C	2025-04-23	2	14.99	29.98
3	4	Gadget D	2025-04-24	5	9.99	49.95
4	5	Tool E	2025-04-25	4	24.99	99.96
5	6	Tool F	2025-04-26	2	49.99	99.98
6	7	Device G	2025-04-27	1	199.99	199.99
7	8	Device H	2025-04-28	3	89.99	269.97
8	9	Widget I	2025-04-29	6	12.99	77.94
9	10	Gadget J	2025-04-30	7	8.99	62.93

```
In [28]: %load_ext sql
The sql extension is already loaded. To reload it, use:
%reload_ext sql

In [29]: %sql sqlite:///sales_data.db

In [33]: conn = sqlite3.connect('sales_data.db')

In [34]: %load_ext sql
The sql extension is already loaded. To reload it, use:
%reload_ext sql

In [35]: %sql sqlite:///sales_data.db

In [40]: %config SqlMagic.style = 'DEPRECATED_DEFAULT'

In [42]: %sql
SELECT *, COUNT(*) AS total_records
FROM sales
* sqlite:///sales_data.db
Done.

Out[42]:
```

	product_id	product_name	order_date	qty	price	sales	total_records
	1	Widget A	2025-04-21	3	19.99	59.97	10

```
In [43]: %sql
SELECT product_name,
SUM(sales) AS total_sales
FROM sales
GROUP BY product_name
ORDER BY total_sales DESC
* sqlite:///sales_data.db
Done.

Out[43]:
```

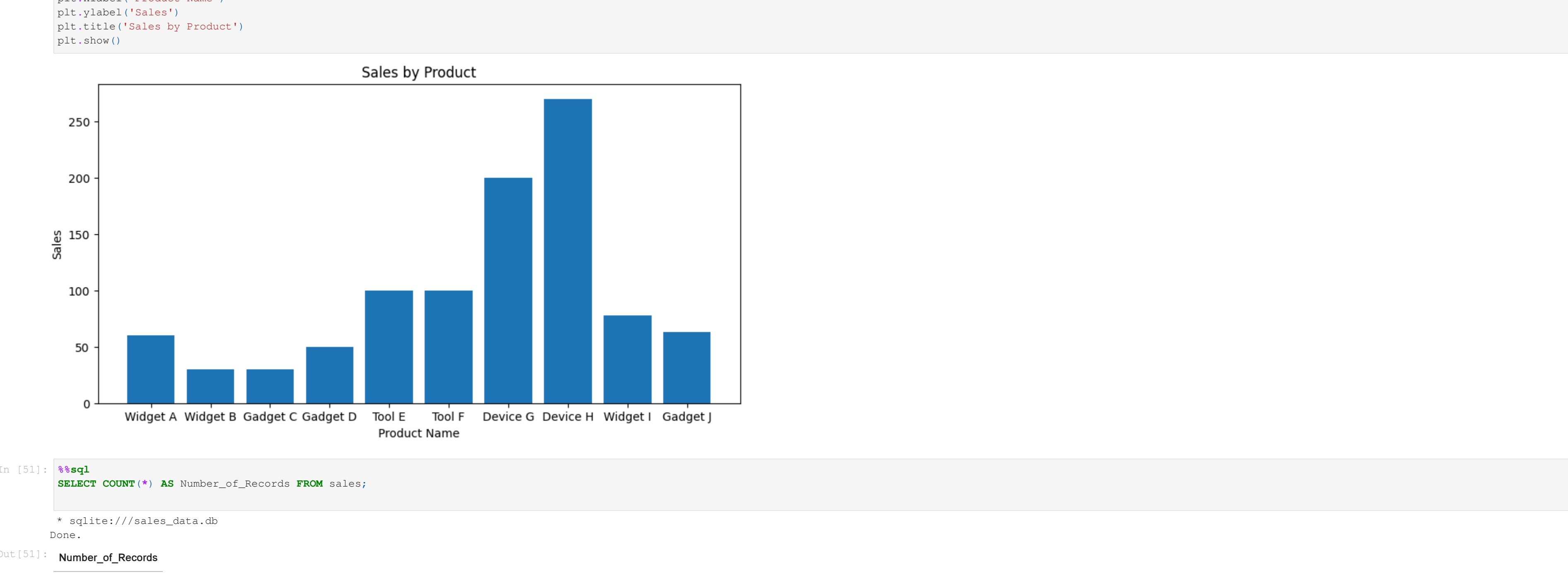
	product_name	total_sales
	Device H	269.97
	Device G	199.99
	Tool F	99.98
	Tool E	99.96
	Widget I	77.94
	Gadget J	62.93
	Widget A	59.97
	Gadget D	49.95
	Widget B	29.99
	Gadget C	29.98

```
In [46]: import sqlite3
import pandas as pd
import matplotlib.pyplot as plt

In [48]: conn = sqlite3.connect('sales_data.db')
cursor = conn.cursor()

In [49]: query = "SELECT product_name, sales FROM sales"
df = pd.read_sql_query(query, conn)

In [50]: # Create the bar chart
plt.figure(figsize=(10, 5))
plt.bar(df['product_name'], df['sales'])
plt.xlabel('Product Name')
plt.ylabel('Sales')
plt.title('Sales by Product')
plt.show()
```



```
In [51]: %sql
SELECT COUNT(*) AS Number_of_Records FROM sales;
* sqlite:///sales_data.db
Done.

Out[51]:
```

	Number_of_Records
	10

```
In [53]: %sql
SELECT product_name,
SUM(sales) AS total_sales
FROM sales
GROUP BY product_name
ORDER BY total_sales;
* sqlite:///sales_data.db
Done.

Out[53]:
```

	product_name	total_sales
	Gadget C	29.98
	Widget B	29.99
	Gadget D	49.95
	Widget A	59.97
	Gadget J	62.93
	Widget I	77.94
	Tool E	99.96
	Tool F	99.98
	Device G	199.99
	Device H	269.97

```
In [54]: %sql
SELECT order_date, SUM(qty) AS total_qty
FROM sales
GROUP BY order_date
ORDER BY order_date;
* sqlite:///sales_data.db
Done.

Out[54]:
```

	order_date	total_qty
	2025-04-21	3
	2025-04-22	1
	2025-04-23	2
	2025-04-24	5
	2025-04-25	4
	2025-04-26	2
	2025-04-27	1
	2025-04-28	3
	2025-04-29	6
	2025-04-30	7

```
In [56]: import matplotlib.pyplot as plt
import seaborn as sns

In [61]: import sqlite3
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

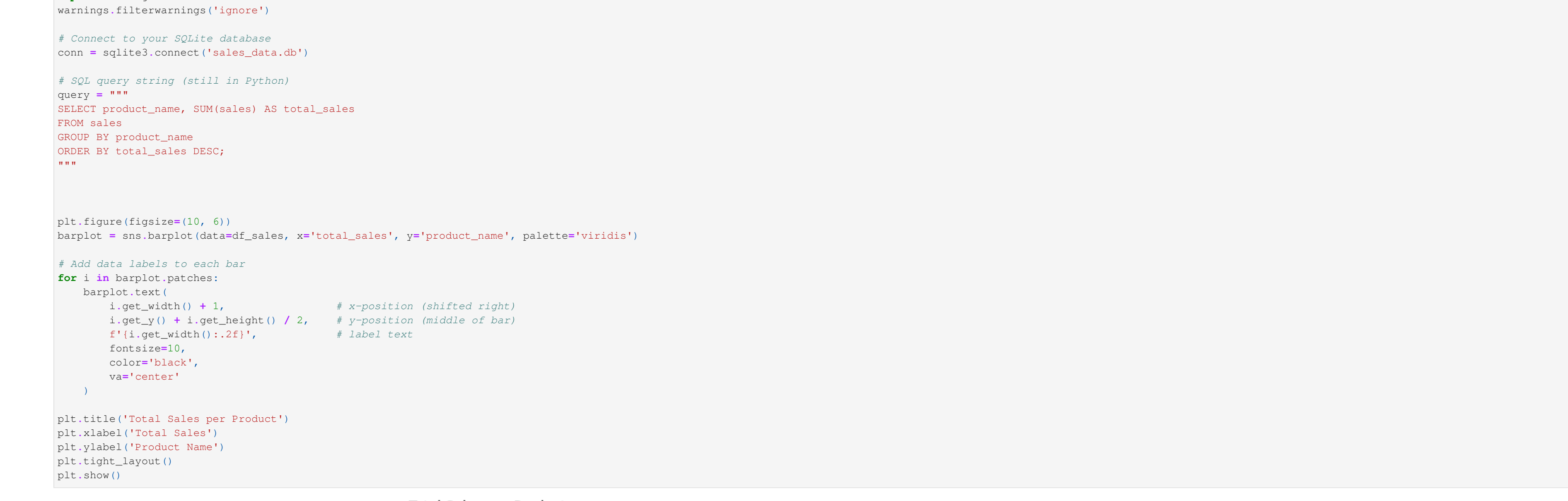
# Connect to your SQLite database
conn = sqlite3.connect('sales_data.db')

# SQL query string (still in Python)
query = """
SELECT product_name, SUM(sales) AS total_sales
FROM sales
GROUP BY product_name
ORDER BY total_sales DESC;
"""

plt.figure(figsize=(10, 6))
barplot = sns.barplot(data=df_sales, x='total_sales', y='product_name', palette='viridis')

# Add data labels to each bar
for i in barplot.patches:
    barplot.text(
        i.get_width() + 1, # x-position (shifted right)
        i.get_y() + i.get_height() / 2, # y-position (middle of bar)
        f'{i.get_width():.2f}', # label text
        fontsize=10,
        color='black',
        va='center'
    )

plt.title('Total Sales per Product')
plt.xlabel('Total Sales')
plt.ylabel('Product Name')
plt.tight_layout()
plt.show()
```



```
In [82]: # SQL query to get total sales per day
query = """
SELECT order_date, SUM(sales) AS total_sales
FROM sales
GROUP BY order_date
ORDER BY order_date;
"""

# Load into DataFrame
df_daily_sales = pd.read_sql_query(query, conn)

# Convert order_date to datetime
df_daily_sales['order_date'] = pd.to_datetime(df_daily_sales['order_date'])

# Plot line chart
plt.figure(figsize=(10, 6))
sns.lineplot(data=df_daily_sales, x='order_date', y='total_sales', marker='o')
plt.title('Total Daily Sales Over Time')
plt.xlabel('Order Date')
plt.ylabel('Total Sales')
plt.grid(True)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



